

Design and Analysis of Algorithm (DAA), Lab-06

BTech (CS-B and CE), 3rd Semester
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Questions

1. **1D array operations and their complexities:** Given an array with n unsorted integer elements, what is the worst-case complexity of the following problems in terms of n :

- (i) Finding the maximum element.
- (ii) Finding the first and second largest elements.
- (iii) Finding the mean.
- (iv) Finding the median.
- (v) Finding the standard deviation.
- (vi) Finding the mode.
- (vii) Removing all duplicates.
- (viii) Reversing the elements of the array.
- (ix) Partitioning the array with respect to a given random pivot element so that all elements less than the pivot appear *after* all the elements greater than or equal to the pivot.

By choosing the proper input representation, write a program in C to validate your procedures and derive the corresponding complexity analysis.

2. **2D square matrix operations and their complexities:** Given square matrices with n rows and n columns, what is the worst-case computational complexity of the following matrix algorithms in terms of n :

- (i) Matrix Addition.
- (ii) Matrix Multiplication.
- (iii) Finding if the given matrix is a zero matrix.
- (iv) Finding if the given matrix is a symmetric matrix.
- (v) Computing the determinant of the given matrix.
- (vi) Transposing the matrix in situ (in place).
- (vii) Finding the eigenvalue and eigenvector of the matrix.

By choosing the proper input representation, write a program in C to validate your procedures and derive the corresponding complexity analysis.

3. **Convolution operation on vectors of size n :** The convolution of two vectors A and B is a new vector C such that

$$C[k] = \sum_{j=0}^{m-1} A[j]B[k-j]$$

Assuming the lengths of A and B are of length m and n , respectively, and the indexing of the vectors starts from 0, find an $O(n \log n)$ divide and conquer algorithm for this operation. Consider, $n \geq m$ for your implementation and analysis.

By choosing the proper input representation, write a program in C to validate your algorithm.

4. **Sorting via reversal procedure:** Suppose you are given a permutation p of the integers 1 to n , and seek to sort them to be in increasing order $[1, \dots, n]$. The only operation at your disposal is **reverse**(**p**, **i**, **j**), which reverses the elements of a subsequence $p_i, \dots, p_j$ in the permutation. For example, the permutation $[1, 4, 3, 2, 5]$ one reversal (of the second through fourth elements) suffices to sort.

- Mathematically, show that it is possible to sort any permutation using $O(n)$ reversals.
- Now suppose that the cost of **reverse**(**p**, **i**, **j**) is equal to its length, the number of elements in the range, $|j - i| + 1$. Design an algorithm that sorts p in $O(n \log^2 n)$ cost. Analyse the running time and cost of your algorithm and prove the correctness.

Write a program in C to validate your algorithm supporting your analysis.
